

Clinical biochemical investigations in the GI system

Fordøjelsessystem, patofysiologi og diagnostik

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Læringsmål (fra studieordning)

- Forstå principperne bag biokemiske metoder for undersøgelse af funktion i fordøjelsessystemet, samt tolkning og mulige anvendelse i klinikken
- Redegøre for betydningen af biologisk og analytisk nøjsikkerhed samt begreber som sensitivitet, specificitet og prædiktive værdier
- Redegøre for sandsynlige årsager til icterus og blødninger i mave-tarmkanalen, baseret på anamnese, objektiv undersøgelse og biokemiske undersøgelser
- Analysere og integrere kliniske og parakliniske parametre som en del af diagnostisk udredning

How is a biochemical lab test evaluated?

A serum sample reveals an ALAT concentration of 120 U/L. What is the diagnosis?

1. Hepatitis
2. Drug-induced liver injury
3. Alcohol-related liver disease
4. A disease not related to the liver



How is a biochemical lab test evaluated?

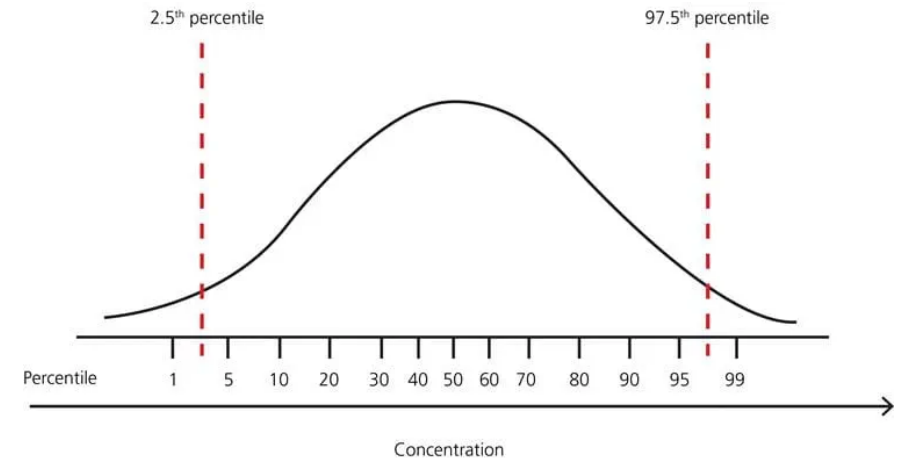
A single lab test can hardly provide the diagnosis

Points to consider:

- Biological variation between patients
- Analytical variation (quantitative assays)
- Sensitivity and specificity (qualitative assays)
- Clinical context (predictive values)

Biological variation: reference intervals

- Defining a single normal value for a laboratory test is unrealistic
- Reference intervals: established by measuring a large population of healthy individuals.
- The central 95% of these values are considered the normal range



Factors influencing biological variation

Age

Sex

Circadian rhythm

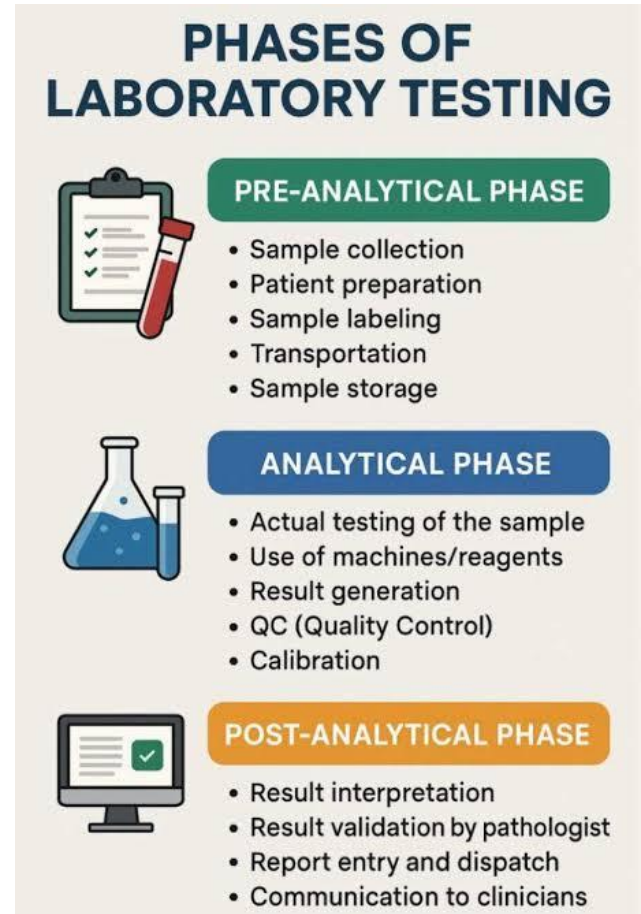
Diet

Genetics

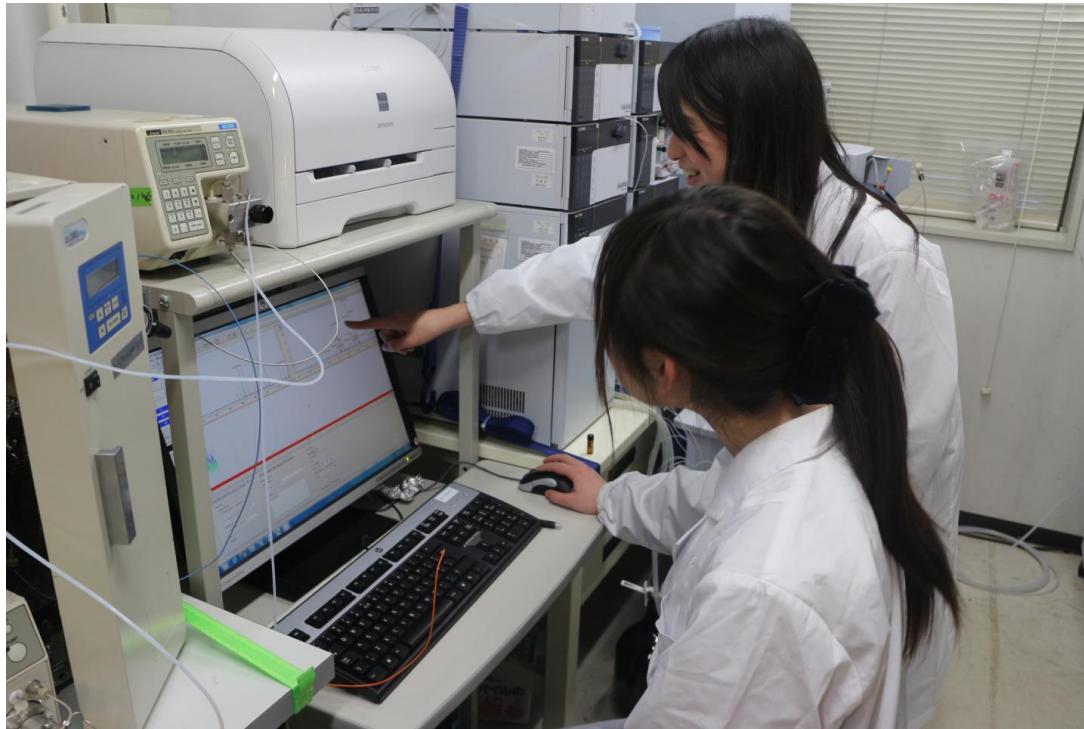
Analytical variation

- Errors can be introduced in different phases
- **Analytical variation** is caused by variations in the measurement technique
 - Instrument
 - Reagents
 - Calibration
 - Technical error

How do we take analytical variations in account?



Analytical methods



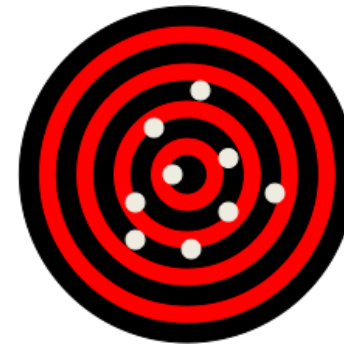
- Analytical methods typically involve measurement of the **quantitative** concentration of a given component (so called analyte) on a biological sample
- Analytical methods can be characterized by a number of performance indicators:
 - Precision
 - Accuracy
 - Analytical sensitivity

Analytical methods

- **Precision:** refers to the reproducibility of analytical results. When a result is precise, numerical results agree closely.
- **Accuracy:** describes the correctness of a result by its closeness to an accepted or 'true' value.



Precise, not accurate



Accurate, not precise



Accurate and precise

Analytical methods

- Precision is usually expressed either as one standard deviation of the mean or as coefficient of variation of the mean
- Example: measurements of fasting serum glucose from a patient

Measurement (mM)
2.2
2.3
2.5
2.5
2.6

average = 2.42 mM

sd = 0.165 mM

*coefficient of variation = $\frac{sd * 100\%}{avg} = 6.82\%$*



Analytical methods

- Accuracy is usually expressed in terms of confidence intervals, which provide the range of values where there is a given probability that the 'true' value lies

- Example: same measurements as before

1. Determine the degrees of freedom

$$\text{degrees of freedom} = n - 1 = 4$$

2. Find the Student's t value according to the desired confidence interval

3. Determine the accuracy, expressed as a confidence interval

Table 1.7 **Values of Student's t**

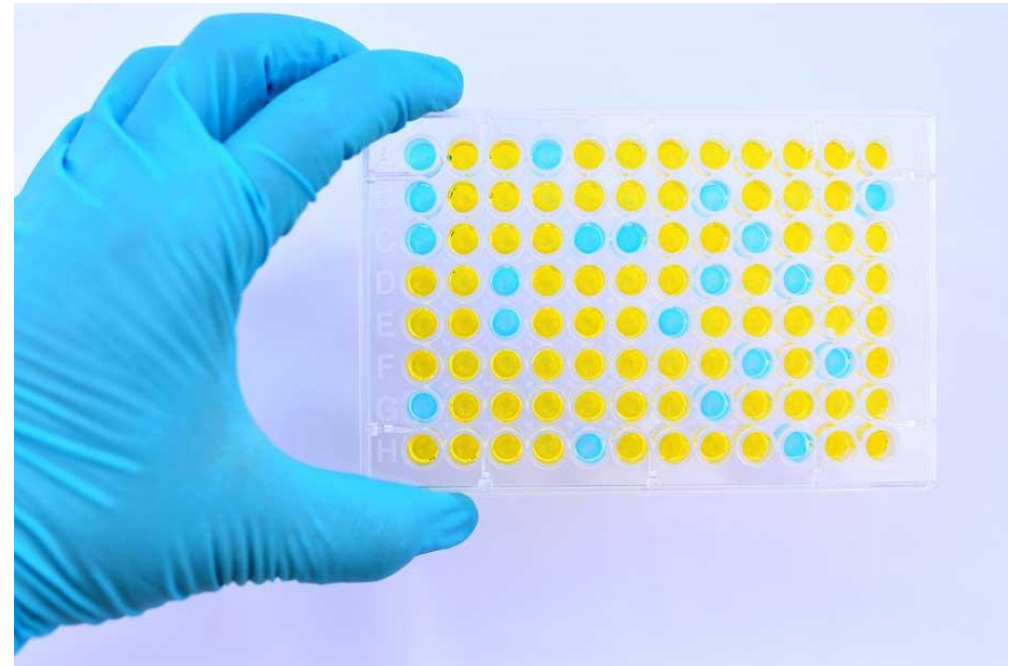
Degrees of freedom	Confidence level (%)		
	50	90	95
2	0.816	2.920	4.303
3	0.765	2.353	3.182
4	0.741	2.132	2.776
5	0.727	2.015	2.571

For 95% confidence interval is $t=2.776$

$$\begin{aligned}\text{confidence interval} &= \text{avg} \pm \frac{t \cdot sd}{\sqrt{n}} \\ &= 2.42 \pm \frac{2.776 \cdot 0.165}{\sqrt{5}} \\ &= 2.42 \pm 0.20 \text{ mM}\end{aligned}$$

Qualitative assays

- In qualitative assays the results are not reported as an exact numeric measurement
- Example: A test whose results are reported as either positive or negative



Qualitative ELISA for detection of HAV antibodies in serum
Yellow=positive / Blue = negative

Qualitative assays

- **Sensitivity** is the ability of the test to detect the target disorder without false positives

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **Specificity** is the ability of the method to give a consistent negative result for known negatives

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

		Disease:	
		Sick	Healthy
Test result:	Positive	True positive (TP)	False positive (FP)
	Negative	False negative (FN)	True negative (TN)
		↓ Sensitivity	↓ Specificity

Qualitative assays

Scenario:

A new rapid diagnostic test for Hepatitis C antibody detection is evaluated on a cohort of 200 clinical samples (100 confirmed positive cases and 100 confirmed healthy controls).

Assay Results:

- Out of 100 positive cases, the test detects 92 (8 are missed).
- Out of 100 negative controls, the test correctly identifies 95 as negative (5 give a false signal).
- Determine the sensitivity and specificity of the assay

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$



Qualitative assays

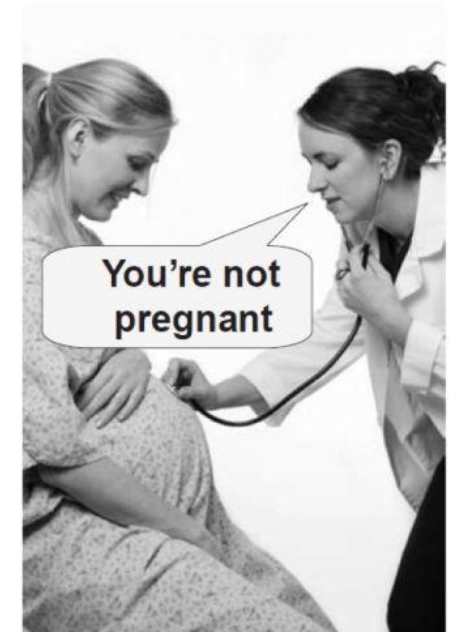
- **Type I Error (α / False Positive):**
 - *What happened:* Test is **Positive**, but the patient is **Healthy** (FP = 5).
 - Lower specificity leads to more Type I errors.
- **Type II Error (β / False Negative):**
 - *What happened:* Test is **Negative**, but the patient is **Sick** (FN = 8).
 - Lower sensitivity leads to more Type II errors.

What are the clinical impacts of type I and Type II errors?

Type I error
(false positive)



Type II error
(false negative)



Clinical context: predictive values

- A positive test does **not** give a diagnosis but changes the **probability** of disease.
 - Positive predictive value $PPV = \frac{TP}{TP + FP}$
 - Negative predictive value $NPV = \frac{TN}{TN + FN}$
- *Predictive values are influenced by the **prevalence** of the disease in the population being tested*

		Disease:		
		Sick	Healthy	
Test result:	Positive	True positive (TP)	False positive (FP)	→ PPV
	Negative	False negative (FN)	True negative (TN)	→ NPV
		↓ Sensitivity	↓ Specificity	

Clinical context: predictive values

Example: iFOBT in cancer diagnostics

Group of young, healthy subjects

	Cancer	No cancer
Positive	8 (TP)	32 (FP)
Negative	2 (FN)	958 (TN)

Sensitivity:

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{8}{8 + 2} = \frac{8}{10} = 80\%$$

Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{958}{958 + 32} = \frac{958}{990} = 96.8\%$$

PPV:

$$\frac{8}{8 + 32} = 20\%$$

$$NPV = \frac{958}{958 + 2} = \frac{958}{960} = 99.8\%$$

High risk subjects

	Cancer	No cancer
Positive	80 (TP)	40 (FP)
Negative	20 (FN)	860 (TN)

Sensitivity:

$$\text{Sensitivity} = \frac{80}{80 + 20} = \frac{80}{100} = 80\%$$

Specificity:

$$\text{Specificity} = \frac{860}{860 + 40} = \frac{860}{900} = 95.6\%$$

PPV:

$$\frac{80}{80 + 40} = 67\%$$

$$NPV = \frac{860}{860 + 20} = \frac{860}{880} = 97.7\%$$

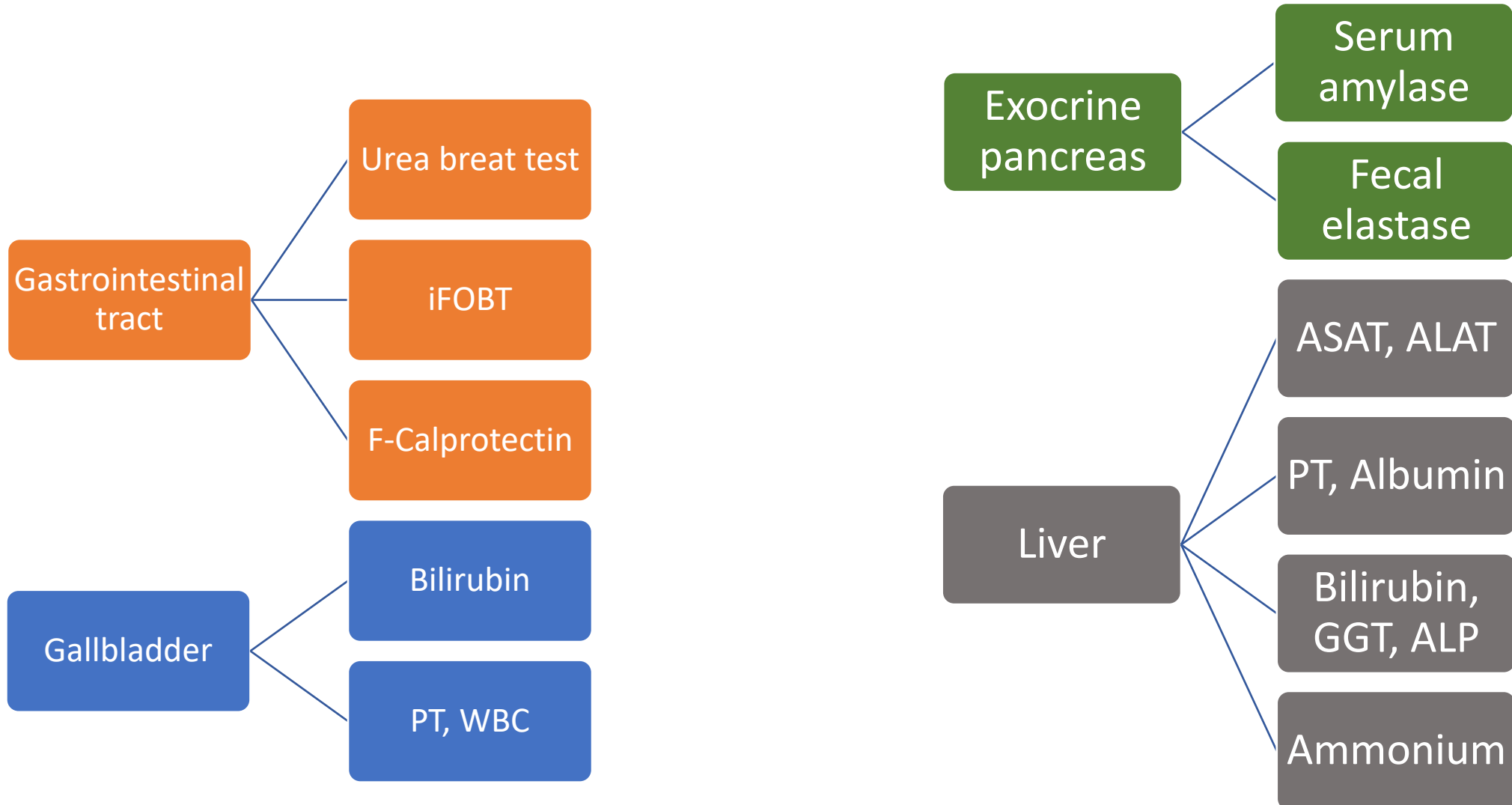
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Clinical biochemical tests of GI function



1. Gastrointestinal tract

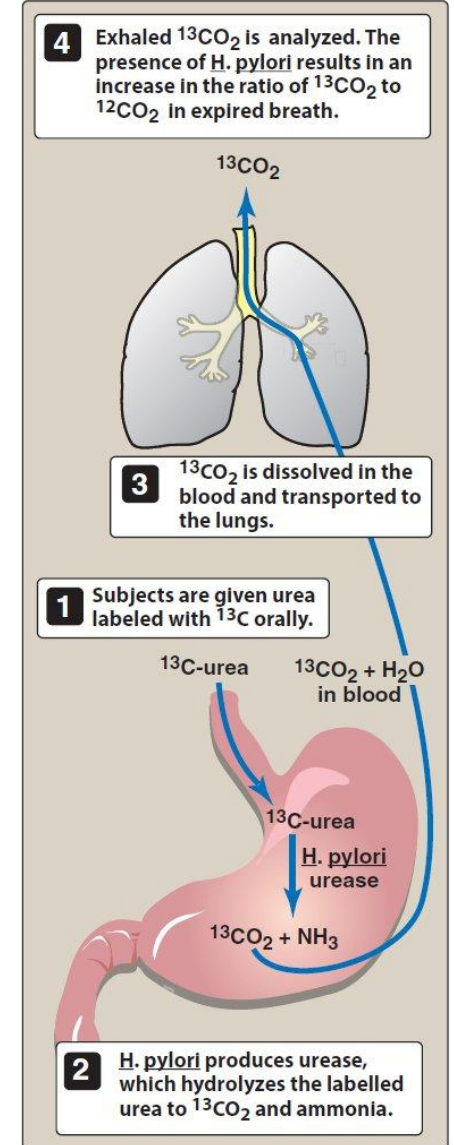
Urea breath test

Indication

- Diagnostic of *Helicobacter Pylori* infection

Principle

- Ingestion of ^{13}C -labeled urea
- Urease will hydrolyze urea into ammonia and carbon dioxide



1. Gastrointestinal tract

iFOBT (immunochemical Fecal Occult Blood Test)

Indication

- Investigation of iron deficiency anemia
- Screening for colorectal cancer

Principle

- Chromogenic detection (guaiac)
- Anti-hemoglobin antibodies applied to a stool sample



1. Gastrointestinal tract

F-Calprotectin

Indication

- Differential diagnosis of IBD
- Monitoring progress of IBD treatment

Principle

- Calprotectin is a protein found mainly in neutrophils
- Presence in stool samples indicates neutrophil migration and infiltration in the intestinal tract



2. Gallbladder

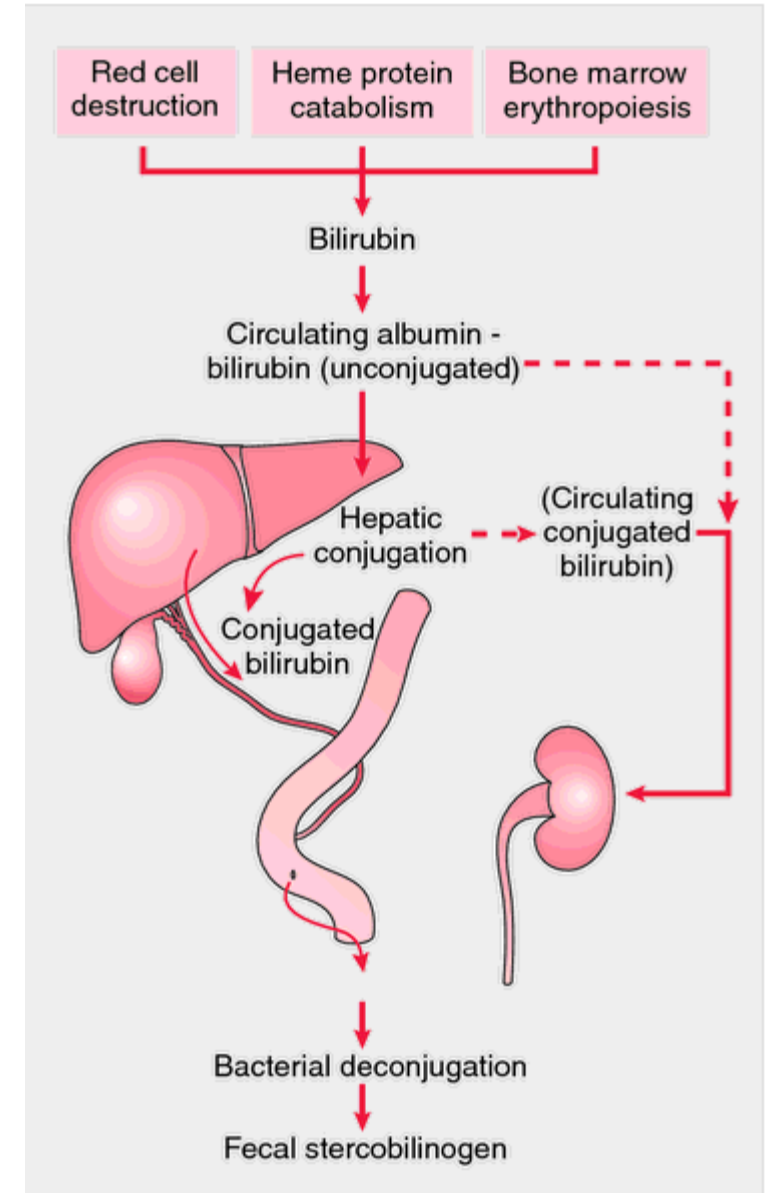
Bilirubin

Indication

- Inflammatory/obstructive diseases of the gallbladder and biliary tract

Assay

- P-concentration of Bilirubin ($\mu\text{mol/L}$)
- Bilirubin in urine (urinstix)



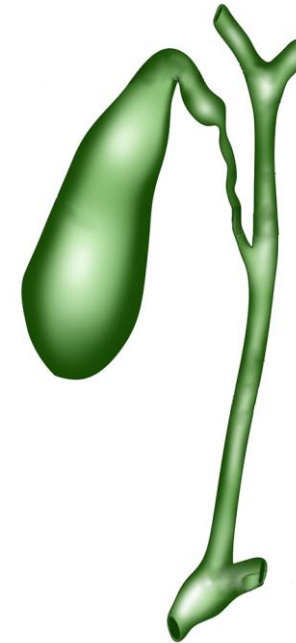
2. Gallbladder

Blood tests

- PT (INR)-> malabsorption of Vit. K
- WBC count -> inflammation

Indication

- Inflammatory/obstructive diseases of the gallbladder and biliary tract



3. Exocrine pancreas

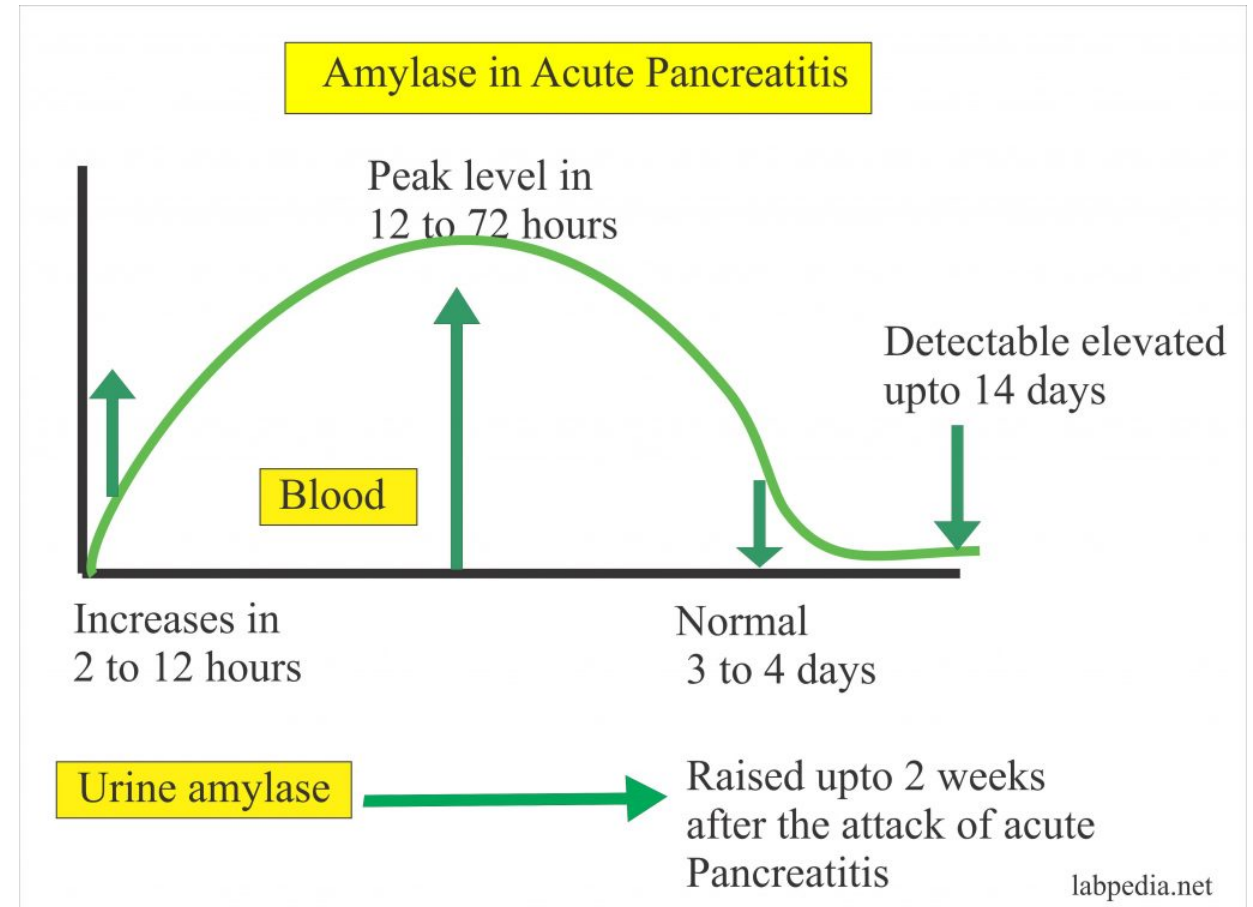
Serum amylase

- P-concentration (U/L)

To assess integrity of pancreas parenchyma

Indication

- Suspicion/control of acute pancreatitis
- Control of chronic pancreatitis



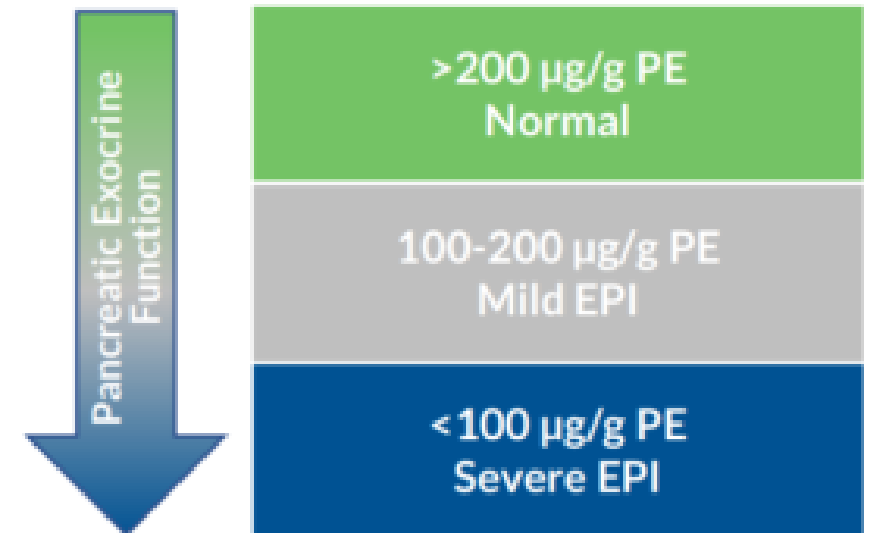
3. Exocrine pancreas

Stool studies

- Fecal Elastase
 - Concentration of pancreas elastase in $\mu\text{g/g}$

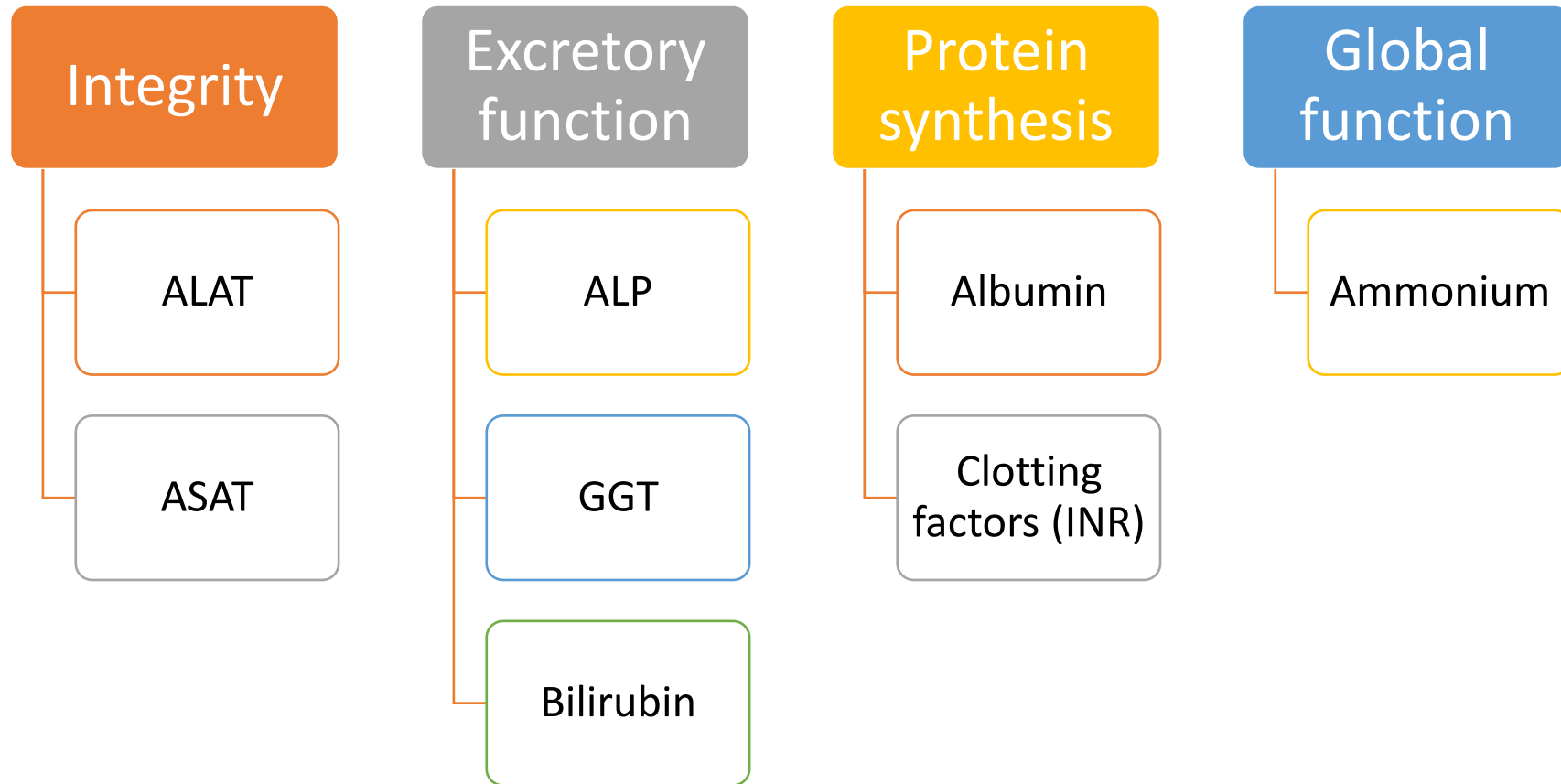
Indication:

- Suspicion/control of acute pancreatitis
- Control of chronic pancreatitis



EPI: exocrine pancreas insufficiency

4. Liver



ALAT, ALP, GGT, bilirubin, albumin and INR -> routine "Levertal"

Liver: Tests for tissue integrity

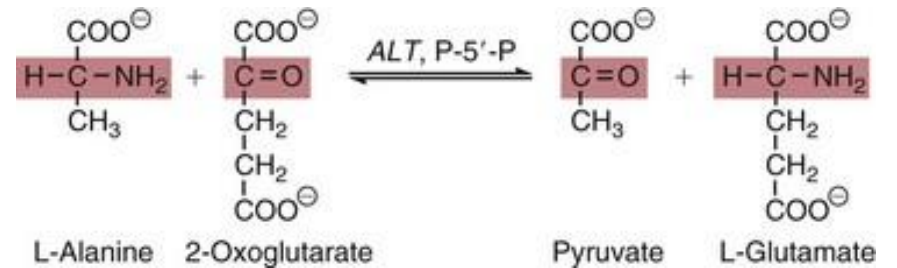
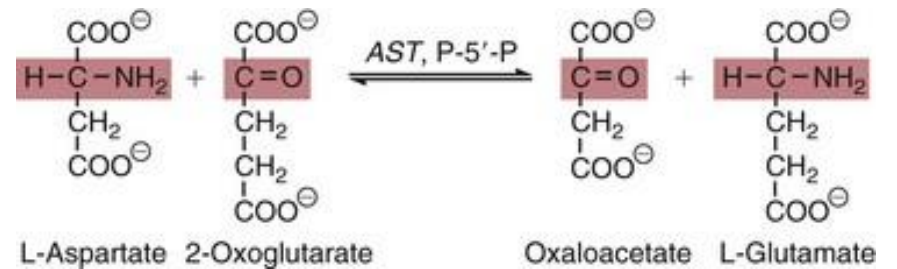
ASpartate AT & ALanine AT

Indicative of acute damage to the liver parenchyma -> release of enzymes

- P-concentration (U/L)

Indication

- Suspicion of liver disease
- Monitoring patients in risk of developing liver disease
- Monitoring patients with known liver disease



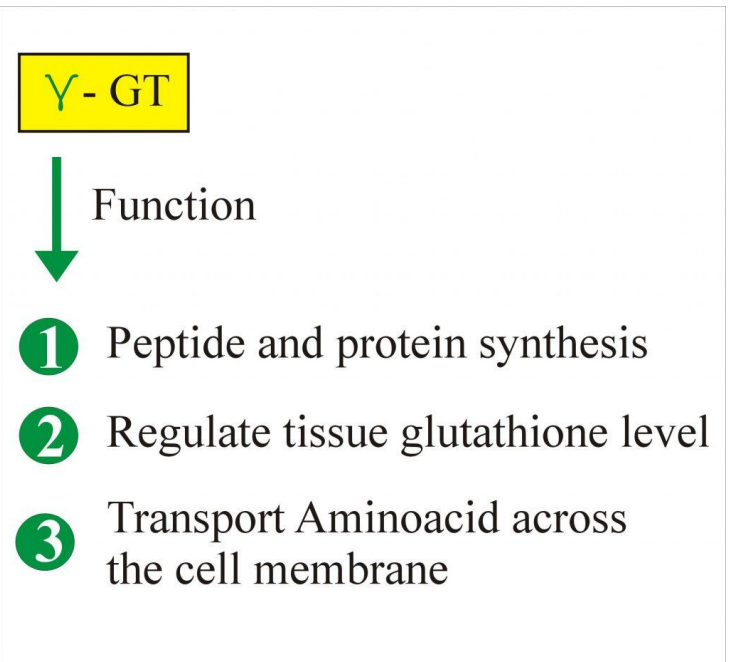
Liver: Tests for excretory function

Alkaline phosphatase, g-glutamyltransferase

- P-concentration ALP & GGT (U/L)
- ALP ->removes the phosphate group from proteins and other molecules
- GGT ->transpeptidase located in the biliary pole of hepatocytes and in cholangiocytes
- ALP & GGT are elevated upon obstruction in the bile ducts

Indication

- Suspicion of liver disease with cholestasis



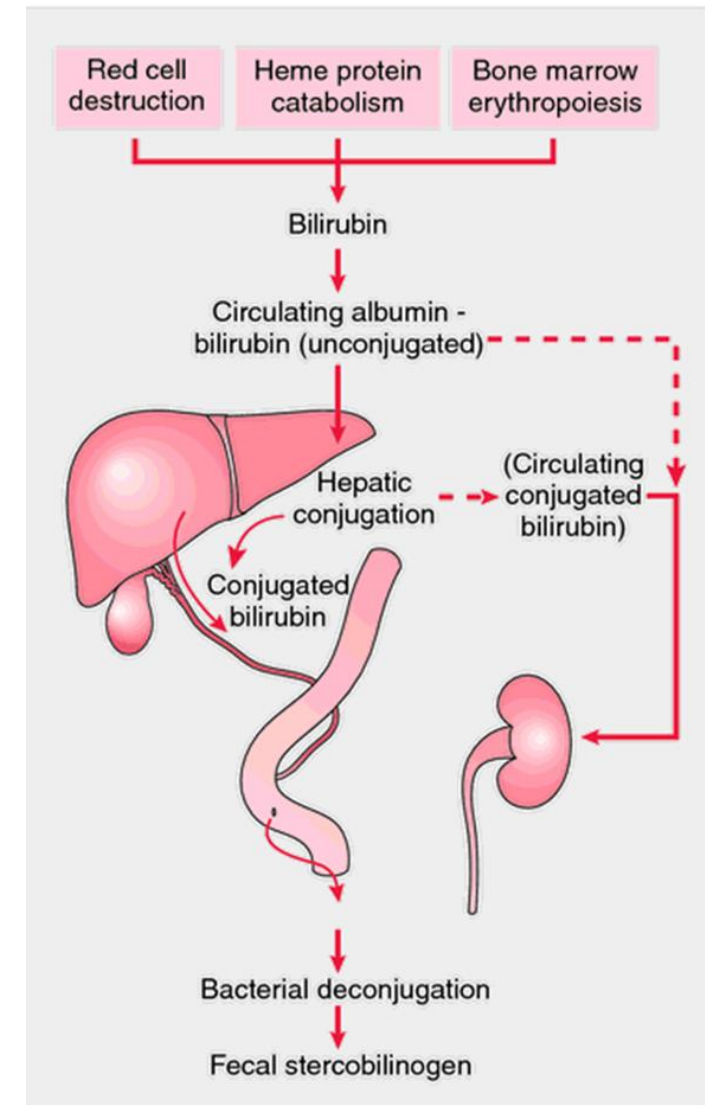
Liver: Tests for excretory function

Bilirubin

- P-concentration of Bilirubin ($\mu\text{mol/L}$)
- Total bilirubin comprises:
 - Unconjugated (80%)
 - Direct (20%)
 - Conjugated bilirubin
 - Delta bilirubin (conj. Bilirubin covalent bound to albumin), present in prolonged cholestatic disease

Indication

- To identify the cause of icterus
- Suspicion of liver disease with cholestasis



Liver: Tests for synthetic function

Prothrombin time - INR

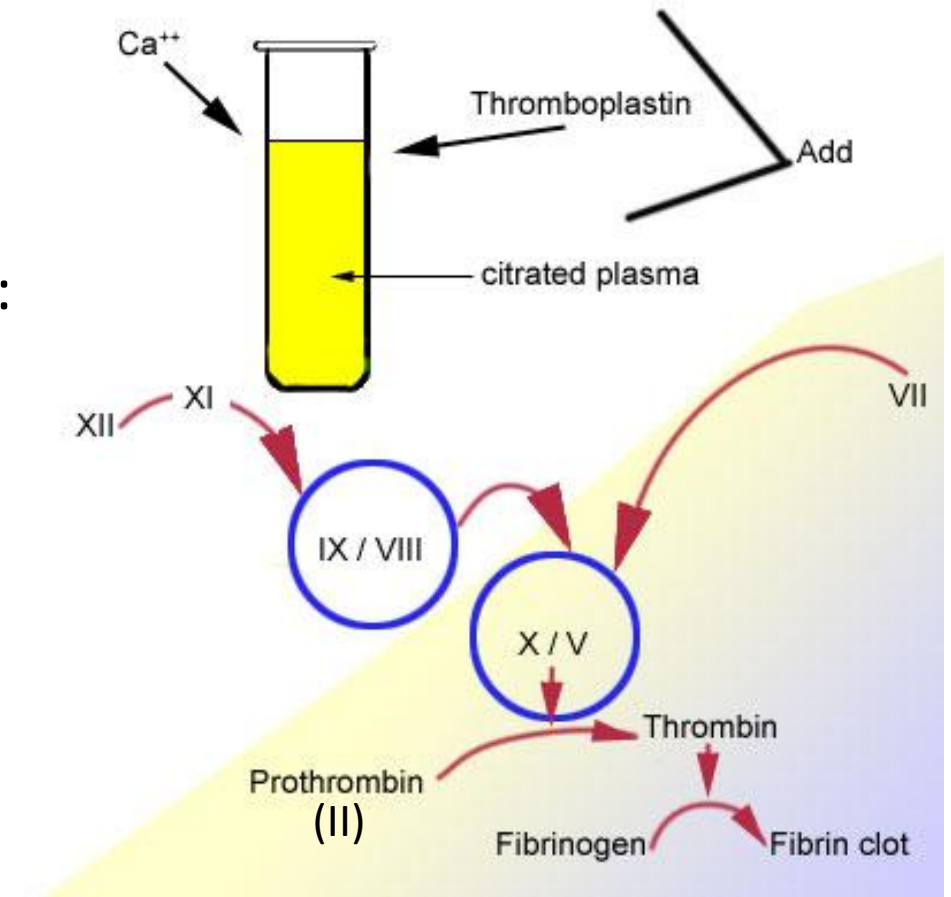
- Concentration of Vit-K dependent coagulation factors affect prothrombin time (PT)
- Expressed as an International Normalized Ratio:

$$INR = \left(\frac{PT_{patient}}{PT_{normal}} \right)^{ISI}$$

ISI=International Sensitivity Index

Indication

- Suspicion of acute or chronic liver disease
- Monitoring chronic liver disease



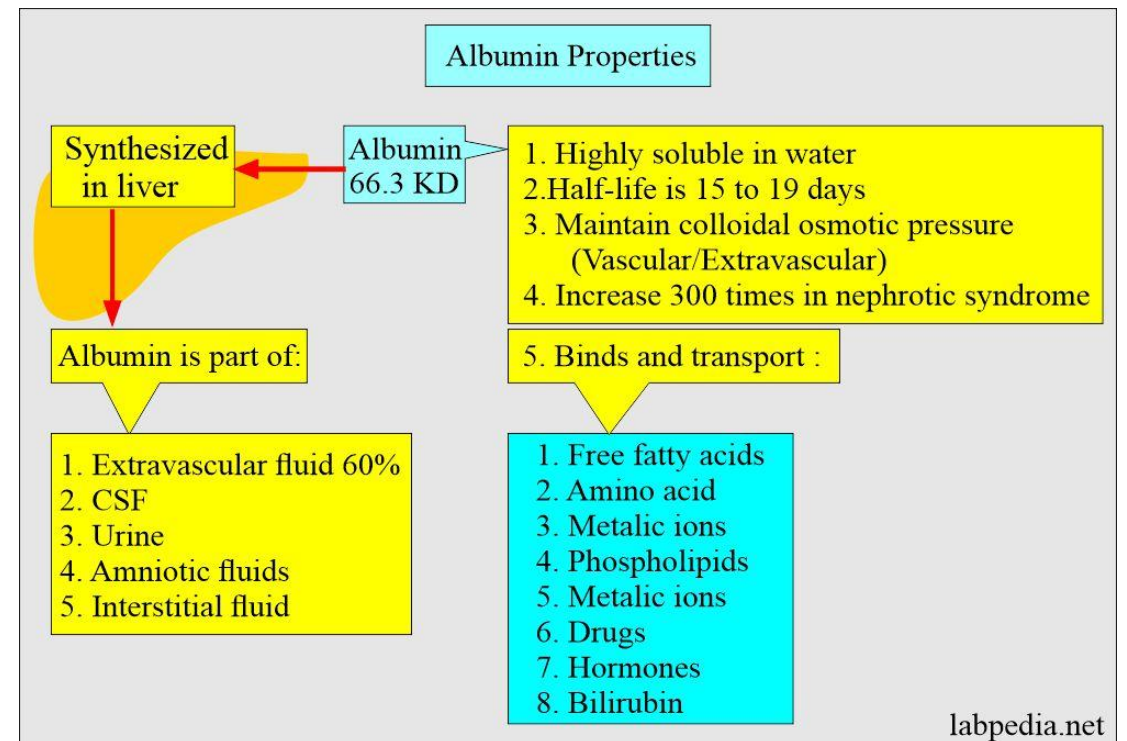
Liver: Tests for synthetic function

Albumin

- Alb P-concentration (g/L or $\mu\text{mol/L}$)
Synthesized exclusively in the liver
Plasma half-life: approx. 20 days

Indication

- Suspicion of chronic liver disease
- Monitoring chronic liver disease



Lab øvelser



Each group receives a serum sample and will carry out:

- 1) Albumin measurement
- 2) Bilirubin measurement

Both methods are based on spectrophotometric determination



Lab øvelser

1) Albumin measurement

- Principle: dye binding method
- Albumin binds with high affinity to Bromocresol purple (BCP) reagent
- The Alb-BCP complex absorbs light at 620 nm
- The absorbance will depend on albumin concentration on the sample

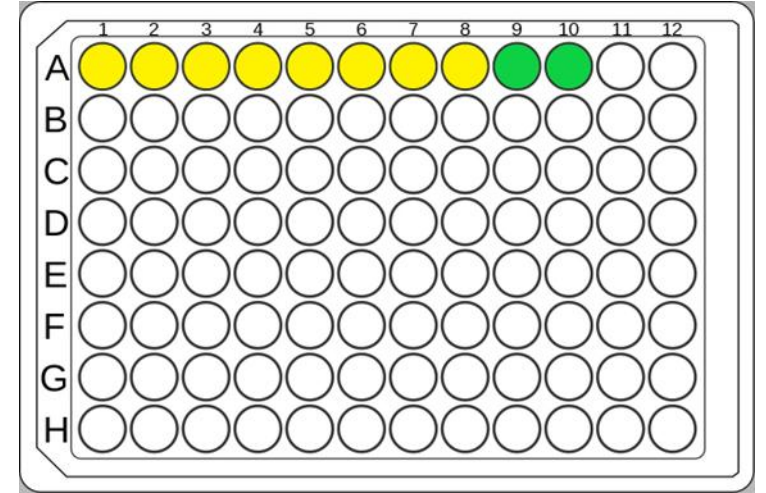
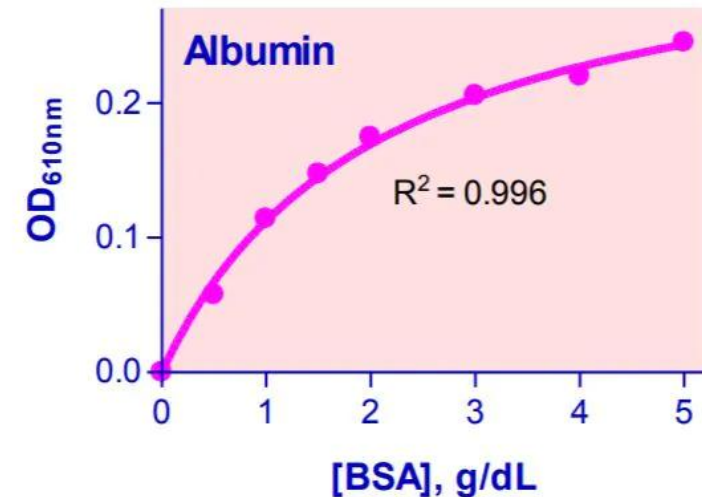


Plate overview: A1-A8= BSA standards; A9-10= Patient's sample



Example of standard curve
BSA=bovine serum albumin

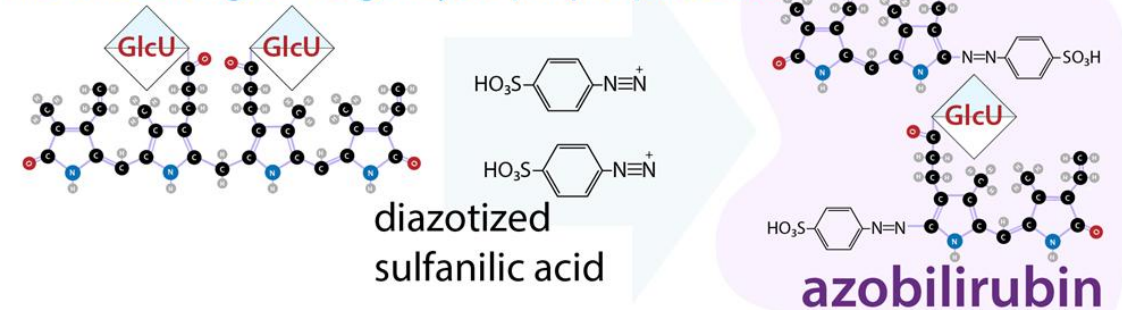
Lab øvelser

2) Bilirubin measurement

- Principle: Jendrassik-Grof or diazo-method.
 - Azobilirubin absorbs light at 540 nm
 - The absorbance will depend on bilirubin concentration
- Steps:
 - I. Direct bilirubin: Conjugated bilirubin reacts with diazo reagent -> azobilirubin
 - II. Total bilirubin: Addition of an accelerator (caffeine benzoate or methanol) dissociates unconjugated bilirubin from albumin
 - III. Unconjugated bilirubin: Total - Direct

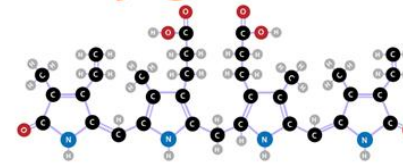
conjugated (direct) bilirubin

this is water soluble so it can react directly with the diazo reagent to give you purple products



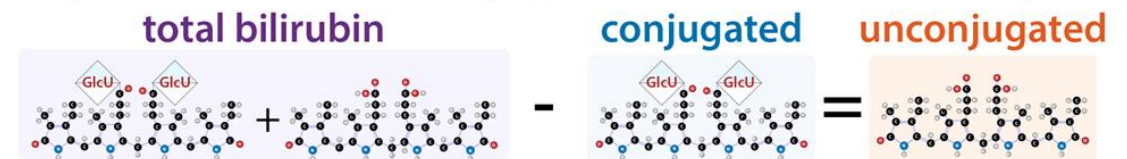
unconjugated (indirect) bilirubin

is not water soluble so you have to add methanol to get it to dissolve & then measure purpleness



that gives you total bilirubin and you subtract the conjugated (direct) bilirubin to indirectly calculate the amount of unconjugated bilirubin

you calculate unconjugated bilirubin indirectly



Lab øvelser

2) Bilirubin measurement

$$\text{Bilirubin total} = \frac{A_{540}(\text{total}) - A_{540}(\text{blank})}{A_{540}(\text{calibrator}) - A_{540}(\text{blank calibrator})} \times \left(5 \frac{\text{mg}}{\text{dL}}\right)$$

$$\text{Bilirubin direct} = \frac{A_{540}(\text{direct}) - A_{540}(\text{blank})}{A_{540}(\text{calibrator}) - A_{540}(\text{blank calibrator})} \times \left(5 \frac{\text{mg}}{\text{dL}}\right)$$

Bilirubin unconjugated = total - direct

Unit conversion: concentration ($\mu\text{mol/L}$) = concentration (mg/dL) x 17,1

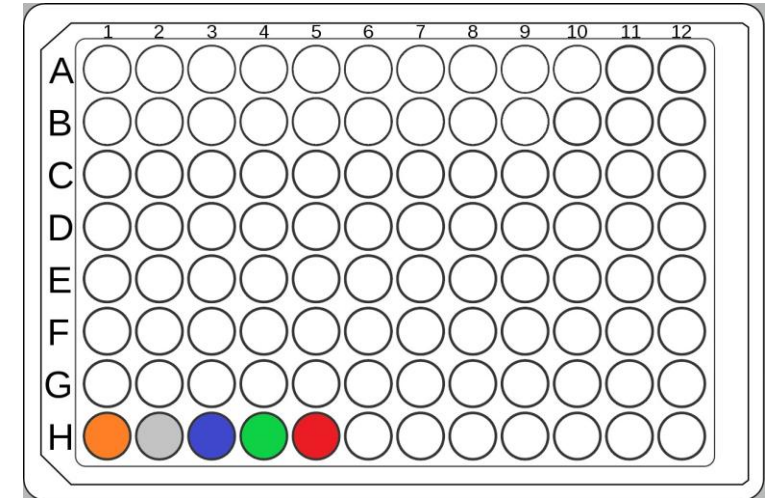


Plate overview: H1= calibrator; H2= DI water;
H3= total; H4= direct; H5= blank

Lab øvelser

Prepare yourself by reading the procedures in moodle



and remember to bring your lab coat!!

